

## MODEL QUESTION PAPER 1

## FIFTH SEMESTER-INSTRUMENTATION ENGINEERING

## COURSE- INDUSTRIAL AUTOMATION

Time: 3 Hour

Max. Marks: 75

**I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)**

|   |                                                                                       | Module Outcomes | Cognitive Level |
|---|---------------------------------------------------------------------------------------|-----------------|-----------------|
| 1 | Define process lag.                                                                   | M1.01           | R               |
| 2 | Split range control configurations has _____measurement and ____manipulated variable. | M1.01           | R               |
| 3 | Write one advantage of DCS                                                            | M2.02           | U               |
| 4 | Define PLC scan time.                                                                 | M2.03           | R               |
| 5 | Write one advantage of LABVIEW.                                                       | M3.02           | U               |
| 6 | Define fuzzy set                                                                      | M3.05           | R               |
| 7 | Give the P & I diagram for the pneumatic control valve                                | M4.05           | R               |
| 8 | Name any one component in safety instrumentation system                               | M4.04           | U               |
| 9 | Give P & I diagram symbol for electric line                                           | M4.05           | R               |

**II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)**

|     |                                                                                             | Module Outcomes | Cognitive Level |
|-----|---------------------------------------------------------------------------------------------|-----------------|-----------------|
| 1.  | Describe compound variable control                                                          | M1.01           | U               |
| 2.  | Illustrate the advantages of feed forward control.                                          | M102            | U               |
| 3.  | Write a short note on Data acquisition system                                               | M2.01           | U               |
| 4.  | List the programming methods of PLC                                                         | M2.04           | U               |
| 5.  | Illustrate the concept of fuzzy logic                                                       | M3.05           | U               |
| 6.  | Write a note on virtual instrumentation.                                                    | M3.01           | U               |
| 7.  | Illustrate the flow line symbols for Pnuematic, electric, hydraulic lines in P & I diagrams | M4.05           | U               |
| 8.  | Describe the classification of hazardous area                                               | M4.01           | U               |
| 9.  | Describe ratio control in brief                                                             | M1.03           | U               |
| 10. | Write a note on the learning methods of ANN.                                                | M3.04           | U               |

**Part – C**

**Answer any six questions. Each question carries 7 marks. (7 × 6 = 42 Marks)**

|      |                                                                | Module Outcomes | Cognitive Level |
|------|----------------------------------------------------------------|-----------------|-----------------|
| III  | Explain cascade control with a neat diagram.                   | M1.02           | U               |
|      | OR                                                             |                 |                 |
| IV   | Illustrate split range control                                 | M1.04           | U               |
| V    | Explain direct digital control with a block diagram            | M2.01           | U               |
|      | OR                                                             |                 |                 |
| VI   | Explain data loggers with a block diagram                      | M2.01           | U               |
| VII  | Compare graphical programming and text based programming       | M3.01           | U               |
|      | OR                                                             |                 |                 |
| VIII | List the features of intelligent control                       | M3.07           | U               |
| IX   | Explain about safety instrumentation systems                   | M4.04           | U               |
|      | OR                                                             |                 |                 |
| X    | Explain Fire and Gas (F & G) system                            | M4.02           | U               |
| XI   | Develop the PLC ladder program for a 4-1 multiplexer.          | M2.04           | U               |
|      | OR                                                             |                 |                 |
| XII  | Describe the features of DCS                                   | M2.02           | U               |
| XIII | Explain the types of instrument protection.                    | M4.02           | U               |
|      | OR                                                             |                 |                 |
| XIV  | Illustrate P&I diagram of a simple distillation column.rement. | M4.05           | U               |

MODEL QUESTION PAPER 2  
**FIFTH SEMESTER- INSTRUMENTATION ENGINEERING**  
**COURSE- INDUSTRIAL AUTOMATION**

Time: 3 Hour

Max. Marks: 75

**I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)**

|   |                                                                | Module Outcomes | Cognitive Level |
|---|----------------------------------------------------------------|-----------------|-----------------|
| 1 | Define process load.                                           | M1.01           | U               |
| 2 | _____control has the theoretical potential for perfect control | M1.02           | R               |
| 3 | Write the full form of SCADA                                   | M2.04           | R               |
| 4 | Define PLC scan time.                                          | M2.03           | R               |
| 5 | Write the full form of LABVIEW.                                | M3.02           | U               |
| 6 | Write one feature of intelligent control.                      | M3.07           | U               |
| 7 | Give the P & I diagram for the pneumatic control valve.        | M4.05           | R               |
| 8 | Write one feature of safety instrumentation system             | M4.04           | U               |
| 9 | Name one method of learning in ANN                             | M2.03           | U               |

**II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)**

|     |                                                                                             | Module Outcomes | Cognitive Level |
|-----|---------------------------------------------------------------------------------------------|-----------------|-----------------|
| 1.  | scribe interactive variable control                                                         | M1.04           | U               |
| 2.  | Illustrate the advantages of cascade control.                                               | M1.03           | U               |
| 3.  | Write short notes on Direct digital control                                                 | M2.01           | U               |
| 4.  | Write any three advantages of PLC                                                           | M2.03           | U               |
| 5.  | Illustrate the concept of fuzzy logic                                                       | M3.05           | U               |
| 6.  | Describe the features of graphical programming.                                             | M3.02           | U               |
| 7.  | Illustrate the flow line symbols for Pneumatic, electric, hydraulic lines in P & I diagrams | M4.05           | R               |
| 8.  | Describe any one type of instrument protection                                              | M4.02           | U               |
| 9.  | Write three features of Direct digital control                                              | M2.01           | U               |
| 10. | Describe independent variable control                                                       | M1.03           | U               |

**Part – C**

**Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)**

|      |                                                              | Module Outcomes | Cognitive Level |
|------|--------------------------------------------------------------|-----------------|-----------------|
| III  | Explain ratio control with a neat diagram.                   | M1.03           | U               |
|      | OR                                                           |                 |                 |
| IV   | Illustrate cascade control                                   | M1.03           | U               |
| V    | Explain the features of DCS                                  | M2.01           | U               |
|      | OR                                                           |                 |                 |
| VI   | Explain supervisory control with a block diagram             | M2.01           | U               |
| VII  | List the advantages of LABVIEW                               | M3.02           | U               |
|      | OR                                                           |                 |                 |
| VIII | Explain fuzzy logic controller                               | M3.05           | U               |
| IX   | Explain about safety instrumentation systems                 | M4.04           | U               |
|      | OR                                                           |                 |                 |
| X    | Explain the selection of equipment based on hazardous area   | M4.03           | U               |
| XI   | Explain the PLC ladder program for a DOL starter.            | M2.04           | U               |
|      | OR                                                           |                 |                 |
| XII  | Explain centralized control with a block diagram             | M2.01           | U               |
| XIII | List the features of intelligent control                     | M3.07           | U               |
|      | OR                                                           |                 |                 |
| XIV  | Describe the learning methods of artificial neural networks. | M3.04           | U               |